

READING PASSAGE 1

Artificial Intelligence in Healthcare

A The study of artificial intelligence in healthcare has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of artificial intelligence in healthcare represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.
- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 4–5

Choose the correct letter, A, B, C or D.

- 4 According to the text, what role do structural factors play?
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations
- 5 What does the passage identify as a significant challenge to current understanding?
 - A. inconsistent results across cultures
 - B. a consistent pattern across diverse contexts
 - C. no measurable effects
 - D. effects limited to Western countries

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.

Question 7

Choose the correct letter, A, B, C or D.

- 7 What finding emerged from the landmark study discussed in the passage?
 - A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 8

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

8 Meta-analyses have confirmed the reliability of the core findings in this field.

Questions 9–10

Choose the correct letter, A, B, C or D.

- 9 The passage indicates that critics have raised concerns about:
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone
- 10 According to the passage, what is one of the key factors identified by researchers?
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone

Questions 11–12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 11 Early researchers had access to the same technologies as modern scholars.
- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 13 Case studies have documented the effects of _____ change on the system.